The Itô Formula: Intuition, Rules, and Applications

References

Main references

- S. Shreve, *Stochastic Calculus for Finance II*, Chapter 4.
- I. Karatzas and S. Shreve, *Brownian Motion and Stochastic Calculus*, chapter on stochastic integration and Itô's rule.
- T. Björk, *Arbitrage Theory in Continuous Time*, chapters on stochastic calculus and stochastic differential equations.

Scope of this lecture

We learn the second-order chain rule, use it to solve basic SDEs, and connect it to products, correlation, discounting, and hedging.

Roadmap

1. See why the ordinary chain rule breaks down for a random path, then turn quadratic variation into a small multiplication table.
2. Understand and apply the Itô formula for Brownian motion.
3. Extend the rule to general one-dimensional Itô processes.
4. Solve GBM and OU/Vasicek, then obtain the mean of a CIR process.
5. Handle products, ratios, and correlated risk factors.
6. Finish with discounting, convexity, and delta-hedging intuition.

Quadratic Variation Becomes a Calculus

Why do we even need a new chain rule?

In finance we constantly want to know how a value that depends on the stock, $V_t = v(t, S_t)$, changes as S_t moves.

The tool we already know

Ordinary calculus says

$$\mathrm{d}v(t, S_t) = v_t \, \mathrm{d}t + v_s \, \mathrm{d}S_t.$$

The question for today

Stock prices do not move along smooth, differentiable paths. Does this formula still hold when S_t is driven by Brownian motion?

Let's just try it, and see what happens

Take the simplest possible case, $f(x) = x^2$ applied to Brownian motion itself. If the ordinary chain rule were correct, we would guess

$$\mathrm{d}(W_t^2) \stackrel{?}{=} 2W_t\,\mathrm{d}W_t, \qquad \text{i.e.} \qquad W_T^2 \stackrel{?}{=} 2\int_0^T W_t\,\mathrm{d}W_t.$$

Your turn

The right-hand side is a stochastic integral with mean zero. Take expectations on both sides. Does the result agree with what you already know about $\mathbb{E}[W_T^2]$?

The naive guess is simply wrong

Taking expectations of the naive guess gives

$$\mathbb{E}[W_T^2] \stackrel{?}{=} 2\,\mathbb{E}\left[\int_0^T W_t \,\mathrm{d}W_t\right] = 0.$$

But we already know

$$\boxed{\mathbb{E}[W_T^2] = T \neq 0.}$$

Something is missing

The ordinary chain rule cannot be correct for Brownian motion. An entire term of order T has been dropped.

Where a smooth path would give zero, Brownian motion does not

For a smooth, differentiable path $x(t)$, a small move satisfies $\Delta x \approx x'(t)\Delta t$, so $(\Delta x)^2$ is of order $(\Delta t)^2$: negligible after summation.

Brownian motion is rougher

Brownian increments scale like

$$\Delta W \sim \sqrt{\Delta t} \quad \Longrightarrow \quad (\Delta W)^2 \sim \Delta t.$$

Squared increments of a smooth path vanish in the limit; squared increments of Brownian motion do not. That is the entire source of the missing term.

A quick numerical check

Split $[0, T]$ into n equal steps and add up the squared increments $\sum_i (\Delta W_i)^2$. Simulating this on a computer for growing n gives, for $T = 1$:

n	10	1,000	100,000
$\sum_i (\Delta W_i)^2$	≈ 0.8	≈ 0.97	≈ 1.00

The pattern

Regardless of the random path drawn, the sum keeps converging to $T = 1$. This limit is exactly the quadratic variation $[W]_T$ from Lecture 2.

This is why we revisit quadratic variation today

We are not repeating Lecture 2 for its own sake. We now know precisely why it matters:

1. the ordinary chain rule silently assumes squared increments vanish;
2. for Brownian motion they do not — they accumulate to $[W]_T = T$;
3. so the chain rule needs a correction term built from quadratic variation.

Today's goal

Turn "squared increments accumulate to $\mathrm{d}t$" into a systematic bookkeeping rule, then use it to fix the chain rule once and for all.

One identity from last lecture contains the new calculus

Left-point sums and Brownian quadratic variation gave

$$\int_0^T W_t \, \mathrm{d}W_t = \frac{1}{2}(W_T^2 - T).$$

Rearranging,

$$\boxed{W_T^2 = 2\int_0^T W_t \, \mathrm{d}W_t + T.}$$

Ordinary-looking part

The term $2W_t \, \mathrm{d}W_t$ is what the classical chain rule predicts.

New part

The extra T comes from $[W]_T = T$.

Today we turn that correction into a systematic rule.

The classical chain rule misses an order-one term

On one small interval,

$$(W_t + \Delta W)^2 - W_t^2 = 2W_t \Delta W + (\Delta W)^2.$$

For a smooth path, the square of the increment disappears after summation. For Brownian motion,

$$\Delta W \sim \sqrt{\Delta t} \quad \Longrightarrow \quad (\Delta W)^2 \sim \Delta t.$$

Thus n small quadratic terms accumulate to a finite correction.

The key change

First-order calculus is not enough: a second-order Taylor term survives.

Orders of magnitude tell us which products survive

On a uniform grid with n steps, $\Delta t = T/n$:

One-step term	Typical size	Sum of n terms	Limit scale
Δt	n^{-1}	1	survives
$(\Delta W)^2$	n^{-1}	1	survives
$\Delta t\, \Delta W$	$n^{-3/2}$	$n^{-1/2}$	vanishes
$(\Delta t)^2$	n^{-2}	n^{-1}	vanishes

Bookkeeping principle

When expanding stochastic differentials, retain terms that accumulate at order $\mathrm{d}t$.

Covariation records accumulated products of increments

For continuous processes X, Y and a partition π, set

$$[X, Y]_t^\pi = \sum_i (X_{t_{i+1}} - X_{t_i})(Y_{t_{i+1}} - Y_{t_i}).$$

When the limit exists as $|\pi| \to 0$, it is denoted by $[X, Y]_t$.

Basic examples

$$[W, W]_t = t, \qquad [W, t]_t = 0, \qquad [t, t]_t = 0.$$

Quadratic variation is the special case $[X]_t = [X, X]_t$.

The one-factor multiplication table

The differential notation summarizes the covariation limits:

$\times$	$\mathrm{d}t$	$\mathrm{d}W_t$
$\mathrm{d}t$	0	0
$\mathrm{d}W_t$	0	$\mathrm{d}t$

Equivalently,

$$(\mathrm{d}t)^2 = 0, \qquad \mathrm{d}t\,\mathrm{d}W_t = 0, \qquad (\mathrm{d}W_t)^2 = \mathrm{d}t.$$

How to read the symbols

The product $\mathrm{d}X_t\,\mathrm{d}Y_t$ is shorthand for the covariation increment $\mathrm{d}[X, Y]_t$. The table is a mnemonic for limits, not literal infinitesimal algebra.

The table with two Brownian shocks includes correlation

Let W and B be Brownian motions with

$$[W, B]_t = \rho t, \qquad -1 \leq \rho \leq 1.$$

Then

$\times$	$\mathrm{d}t$	$\mathrm{d}W_t$	$\mathrm{d}B_t$
$\mathrm{d}t$	0	0	0
$\mathrm{d}W_t$	0	$\mathrm{d}t$	$\rho\,\mathrm{d}t$
$\mathrm{d}B_t$	0	$\rho\,\mathrm{d}t$	$\mathrm{d}t$

For independent Brownian motions, $\rho = 0$ and $\mathrm{d}W_t\,\mathrm{d}B_t = 0$. More generally, $\mathrm{d}W_t^i\,\mathrm{d}W_t^j = \rho_{ij}\,\mathrm{d}t$ for a Brownian vector with correlation matrix (ρ_{ij}).

Application: correlation appears as drift in a product

Apply the discrete product identity in differential form:

$$\mathrm{d}(W_t B_t) = W_t \,\mathrm{d}B_t + B_t \,\mathrm{d}W_t + \mathrm{d}W_t \,\mathrm{d}B_t.$$

Using the table,

$$\boxed{\mathrm{d}(W_t B_t) = W_t \,\mathrm{d}B_t + B_t \,\mathrm{d}W_t + \rho \,\mathrm{d}t.}$$

After integration and expectation,

$$\mathbb{E}[W_t B_t] = \rho t.$$

Sanity check

The stochastic integrals have mean zero under their L^2 conditions, leaving exactly the covariance ρt.

Table drill: same shock, independent shock, correlated shock

Let

$$\mathrm{d}X_t = 2\,\mathrm{d}t + 3\,\mathrm{d}W_t, \quad \mathrm{d}Y_t = -\,\mathrm{d}t + 4\,\mathrm{d}W_t, \quad \mathrm{d}Z_t = -\,\mathrm{d}t + 4\,\mathrm{d}B_t.$$

Your turn

1. Compute $(\mathrm{d}X_t)^2$.
2. Compute $\mathrm{d}X_t\,\mathrm{d}Y_t$: both processes use the same shock.
3. Compute $\mathrm{d}X_t\,\mathrm{d}Z_t$ when W, B are independent.
4. Repeat when $\mathrm{d}W_t\,\mathrm{d}B_t = -\frac{1}{2}\,\mathrm{d}t$.

Predict the sign before doing the last calculation.

Table drill debrief: only diffusion coefficients survive

All terms containing $\mathrm{d}t$ disappear from products of differentials:

$$\boxed{(\mathrm{d}X_t)^2 = 9\,\mathrm{d}t, \qquad \mathrm{d}X_t\,\mathrm{d}Y_t = 12\,\mathrm{d}t, \qquad \mathrm{d}X_t\,\mathrm{d}Z_t = 12\rho\,\mathrm{d}t.}$$

Therefore

$$\mathrm{d}X_t\,\mathrm{d}Z_t = \begin{cases} 0, & \rho = 0, \\ -6\,\mathrm{d}t, & \rho = -\frac{1}{2}. \end{cases}$$

Takeaway

The coefficients $2, -1$ multiplying $\mathrm{d}t$ never enter covariation. Only diffusion loadings and correlation matter.

The Ito Formula Is a Second-Order Chain Rule

Brownian scaling tells us how far Taylor must go

For a smooth function $f(t, x)$, write heuristically

$$\Delta f \approx f_t \Delta t + f_x \Delta W + \frac{1}{2} f_{xx} (\Delta W)^2.$$

The omitted terms have smaller accumulated order:

$$\Delta t \, \Delta W \sim (\Delta t)^{3/2}, \qquad (\Delta W)^3 \sim (\Delta t)^{3/2}.$$

Replacing $(\Delta W)^2$ by Δt suggests

$$\Delta f \approx \left(f_t + \frac{1}{2} f_{xx} \right) \Delta t + f_x \Delta W.$$

The Itô formula for $f(t, W_t)$

Let $f \in C^{1,2}([0, T] \times \mathbb{R})$.

Brownian Itô formula

$$f(t, W_t) = f(0, W_0) + \int_0^t f_t(s, W_s)\,\mathrm{d}s + \int_0^t f_x(s, W_s)\,\mathrm{d}W_s + \frac{1}{2}\int_0^t f_{xx}(s, W_s)\,\mathrm{d}s.$$

In differential notation,

$$\boxed{\mathrm{d}f(t, W_t) = \left(f_t + \tfrac{1}{2} f_{xx}\right)(t, W_t)\,\mathrm{d}t + f_x(t, W_t)\,\mathrm{d}W_t.}$$

The term involving dt is called the drift, while the term involving dW_t is called the diffusion (or volatility) term.

Construction: telescope, expand, and keep surviving scales

On a partition $0 = t_0 < \cdots < t_n = t$, telescope

$$f(t, W_t) - f(0, W_0) = \sum_i \Delta f_i.$$

A second-order Taylor expansion gives

$$\Delta f_i = f_t \Delta t_i + f_x \Delta W_i + \frac{1}{2} f_{xx} (\Delta W_i)^2 + r_i.$$

As the mesh vanishes:

- the first sum becomes $\int f_t \, \mathrm{d}t$;
- the left-point second sum becomes $\int f_x \, \mathrm{d}W$;
- quadratic variation turns the third sum into $\frac{1}{2} \int f_{xx} \, \mathrm{d}t$;
- the accumulated remainder vanishes after localization.

A four-step recipe prevents most mistakes

To compute $\mathrm{d}f(t, W_t)$:

1. Treat t and x as ordinary deterministic variables.
2. Compute $f_t(t,x)$, $f_x(t,x)$, and $f_{xx}(t,x)$.
3. Evaluate the derivatives at $x = W_t$.
4. Assemble the $\mathrm{d}t$ and $\mathrm{d}W_t$ terms separately.

Notation check

Differentiate with respect to x, not "with respect to W_t." Compute first, then substitute $x = W_t$.

Worked examples: powers and a curved transform

For $f(x) = x^2$,

$$\boxed{\mathrm{d}(W_t^2) = 2W_t\,\mathrm{d}W_t + \mathrm{d}t.}$$

Moreover, on $[0, T]$, $\mathbb{E}\int_0^T 4W_s^2\,\mathrm{d}s = 2T^2 < \infty$, so $W_t^2 - t$ is a true martingale.

For $f(x) = x^3$,

$$\boxed{\mathrm{d}(W_t^3) = 3W_t^2\,\mathrm{d}W_t + 3W_t\,\mathrm{d}t.}$$

For $f(x) = \sin x$,

$$\boxed{\mathrm{d}(\sin W_t) = \cos(W_t)\,\mathrm{d}W_t - \tfrac{1}{2}\sin(W_t)\,\mathrm{d}t.}$$

What changed?

The curvature f'' creates drift even though Brownian motion itself has zero drift.

Exercise: manufacture a fourth-order martingale

Apply Itô's formula to $f(x) = x^4$.

Your turn

1. Compute $\mathrm{d}(W_t^4)$.
2. Find c such that
$$W_t^4 - c\int_0^t W_s^2\,\mathrm{d}s$$
has zero drift.
3. Use expectations to recover $\mathbb{E}[W_t^4]$.

Why is the resulting local martingale a true martingale on every bounded horizon?

Exercise debrief: curvature generates the moment recursion

Since $f'(x) = 4x^3$ and $f''(x) = 12x^2$,

$$\mathrm{d}(W_t^4) = 4W_t^3\,\mathrm{d}W_t + 6W_t^2\,\mathrm{d}t.$$

Therefore

$$\boxed{M_t = W_t^4 - 6\int_0^t W_s^2\,\mathrm{d}s}$$

is a true martingale on bounded horizons; the polynomial integrands have all required moments. Taking expectations,

$$\mathbb{E}[W_t^4] = 6\int_0^t \mathbb{E}[W_s^2]\,\mathrm{d}s = 6\int_0^t s\,\mathrm{d}s = 3t^2.$$

Exercise: one more step gives the sixth moment

Fix $T > 0$ and use the result $\mathbb{E}[W_t^4] = 3t^2$.

Your turn

1. Apply Itô's formula to $f(x) = x^6$.
2. Separate the drift and stochastic-integral terms.
3. Take expectations and compute $\mathbb{E}[W_T^6]$.

Why is the stochastic integral a true martingale on every bounded horizon?

Exercise debrief: the moment recursion closes

Since $f'(x) = 6x^5$ and $f''(x) = 30x^4$,

$$\mathrm{d}(W_t^6) = 6W_t^5\,\mathrm{d}W_t + 15W_t^4\,\mathrm{d}t.$$

Because $\mathbb{E}\int_0^T 36W_t^{10}\,\mathrm{d}t < \infty$, the stochastic integral is a true martingale on $[0, T]$. Hence

$$\boxed{\mathbb{E}[W_T^6] = 15\int_0^T \mathbb{E}[W_t^4]\,\mathrm{d}t = 45\int_0^T t^2\,\mathrm{d}t = 15T^3.}$$

More generally, $m_n(t) = \mathbb{E}[W_t^n]$ satisfies

$$m_n'(t) = \frac{n(n-1)}{2}\,m_{n-2}(t).$$

Exercise: exit time and occupation inside an interval

For $a > 0$, let

$$\tau_a = \inf\{t \geq 0 : |W_t| = a\}.$$

Use the martingales

$$W_t^2 - t, \qquad W_t^4 - 6\int_0^t W_s^2 \, \mathrm{d}s.$$

Your turn

Stop first at $t \wedge \tau_a$, then determine

$$\mathbb{E}[\tau_a] \qquad \text{and} \qquad \mathbb{E}\left[\int_0^{\tau_a} W_s^2 \, \mathrm{d}s\right].$$

Why should the limit $t \to \infty$ be taken only after stopping?

Exercise debrief: stopping turns polynomials into identities

Write $\tau_t = t \wedge \tau_a$. Optional sampling gives

$$\mathbb{E}[\tau_t] = \mathbb{E}[W_{\tau_t}^2] \leq a^2.$$

Monotone convergence first yields $\mathbb{E}[\tau_a] \leq a^2$, hence $\tau_a < \infty$ a.s. Then continuity and dominated convergence give

$$\boxed{\mathbb{E}[\tau_a] = a^2.}$$

Stopping the fourth-order martingale similarly gives

$$\mathbb{E}[W_{\tau_t}^4] = 6\mathbb{E}\left[\int_0^{\tau_t} W_s^2 \, \mathrm{d}s\right].$$

Letting $t \to \infty$ now yields

$$\boxed{\mathbb{E}\left[\int_0^{\tau_a} W_s^2 \, \mathrm{d}s\right] = \frac{a^4}{6}.}$$

The corrected exponential has exactly zero drift

Let

$$M_t = \exp\left(\lambda W_t - \frac{1}{2}\lambda^2 t\right).$$

For $f(t,x) = e^{\lambda x - \lambda^2 t/2}$,

$$f_t = -\frac{1}{2}\lambda^2 f, \qquad f_x = \lambda f, \qquad f_{xx} = \lambda^2 f.$$

Thus the two drift terms cancel:

$$\boxed{\mathrm{d}M_t = \lambda M_t \,\mathrm{d}W_t.}$$

Here $M_t \geq 0$ and direct Gaussian calculation gives $\mathbb{E}[M_t] = 1$, so M is a true martingale.

Zero drift is a differential equation for the function

For $Y_t = u(t, W_t)$, the Itô formula gives

$$\mathrm{d}Y_t = \underbrace{\left(u_t + \frac{1}{2}u_{xx}\right)(t, W_t)\,\mathrm{d}t}_{\text{drift}} + u_x(t, W_t)\,\mathrm{d}W_t.$$

Thus the drift vanishes when

$$\boxed{u_t + \frac{1}{2}u_{xx} = 0.}$$

Probabilistic reading

Solutions of the backward heat equation turn Brownian motion into local martingales. Suitable growth or integrability conditions upgrade them to true martingales.

Exercise: which transforms have zero drift?

For fixed $T > 0$ and $\lambda \in \mathbb{R}$, consider

$$u_1(t,x) = x^2 - t, \quad u_2(t,x) = x^2 + t, \quad u_3(t,x) = x^2 + T - t, \quad u_4(t,x) = e^{\lambda x - \lambda^2 t/2}.$$

Your turn

For each function, compute $u_t + \frac{1}{2}u_{xx}$. Decide whether $u_i(t, W_t)$ has zero drift and interpret $u_3(t, W_t)$.

Exercise debrief: the generator gives an instant test

Function	$u_t + \frac{1}{2}u_{xx}$	Conclusion
$x^2 - t$	0	martingale
$x^2 + t$	2	drift $2\,\mathrm{d}t$
$x^2 + T - t$	0	martingale
$e^{\lambda x - \lambda^2 t/2}$	0	martingale

Moreover,

$$W_t^2 + T - t = \mathbb{E}[W_T^2 \mid \mathcal{F}_t].$$

Takeaway

The backward equation encodes a conditional forecast that stays unchanged on average as information arrives.

From Brownian Motion to Ito Processes

An Ito process separates trend from noise

On a fixed horizon $[0, T]$, an *Ito process* has the form

$$X_t = X_0 + \int_0^t a_s \, \mathrm{d}s + \int_0^t b_s \, \mathrm{d}W_s.$$

We assume, for instance,

$$\int_0^T |a_s| \, \mathrm{d}s < \infty \quad \text{a.s.}, \qquad \mathbb{E}\left[\int_0^T b_s^2 \, \mathrm{d}s\right] < \infty,$$

with progressively measurable coefficients a and b.

Term	Role	Size over $\mathrm{d}t$
$a_t \, \mathrm{d}t$	local trend	$\mathrm{d}t$
$b_t \, \mathrm{d}W_t$	local shock	$\sqrt{\mathrm{d}t}$

The shorthand is $\mathrm{d}X_t = a_t \, \mathrm{d}t + b_t \, \mathrm{d}W_t$.

Differential and integral forms describe the same process

The differential form is compact local notation:

$$\boxed{\mathrm{d}X_t = a_t\,\mathrm{d}t + b_t\,\mathrm{d}W_t.}$$

Its precise integral meaning, on every interval $0 \le s \le t \le T$, is

$$\boxed{X_t - X_s = \int_s^t a_u\,\mathrm{d}u + \int_s^t b_u\,\mathrm{d}W_u.}$$

Differential form

Best for local calculations, Itô's formula, and identifying drift and diffusion.

Integral form

Best for stating what the process actually is and comparing values at two times.

Do not divide by $\mathrm{d}t$

Brownian paths are not differentiable: $\mathrm{d}W_t/\mathrm{d}t$ is not an ordinary random function. The differential equation must be understood through its integral form.

Optional, for the curious: the drift and diffusion you read off are the only ones

If you write the same process two different ways,

$$X_t = X_0 + \int_0^t a_s \, \mathrm{d}s + \int_0^t b_s \, \mathrm{d}W_s = X_0 + \int_0^t \widetilde{a}_s \, \mathrm{d}s + \int_0^t \widetilde{b}_s \, \mathrm{d}W_s,$$

it turns out $a = \widetilde{a}$ and $b = \widetilde{b}$ — there is only one correct drift and only one correct diffusion coefficient for a given process. Intuitively: a purely-$\mathrm{d}t$ term (finite variation, zero quadratic variation) and a purely-$\mathrm{d}W$ term (zero drift on average, positive quadratic variation) have such different "signatures" that they can only cancel each other if both are zero.

Why this matters in practice

It means the drift/diffusion split you get after applying Itô's formula is not a matter of algebraic convenience: it is *the* answer, and every later formula (delta hedge, PDE, martingale test) can safely rely on it.

Quadratic variation sees volatility, not drift

If

$$\mathrm{d}X_t = a_t\,\mathrm{d}t + b_t\,\mathrm{d}W_t,$$

then the multiplication table gives

$$(\mathrm{d}X_t)^2 = a_t^2(\mathrm{d}t)^2 + 2a_t b_t\,\mathrm{d}t\,\mathrm{d}W_t + b_t^2(\mathrm{d}W_t)^2 = b_t^2\,\mathrm{d}t.$$

Therefore

$$\boxed{\mathrm{d}[X]_t = b_t^2\,\mathrm{d}t, \qquad [X]_t = \int_0^t b_s^2\,\mathrm{d}s.}$$

Interpretation

Quadratic variation measures accumulated instantaneous variance. A finite-variation drift contributes nothing to it.

The general one-dimensional Ito formula

Let

$$\mathrm{d}X_t = a_t\,\mathrm{d}t + b_t\,\mathrm{d}W_t, \qquad f \in C^{1,2}.$$

Then

$$\boxed{\mathrm{d}f(t,X_t) = \left(f_t + a_t f_x + \frac{1}{2}b_t^2 f_{xx}\right)(t,X_t)\,\mathrm{d}t + b_t f_x(t,X_t)\,\mathrm{d}W_t.}$$

Equivalently,

$$\mathrm{d}f = f_t\,\mathrm{d}t + f_x\,\mathrm{d}X + \frac{1}{2}f_{xx}\,\mathrm{d}[X].$$

What changed from ordinary calculus?

Only the curvature correction $\frac{1}{2}f_{xx}\,\mathrm{d}[X]$. Everything else looks like the usual chain rule.

Worked example: square a general Ito process

Apply $f(x) = x^2$ to $\mathrm{d}X_t = a_t\,\mathrm{d}t + b_t\,\mathrm{d}W_t$:

$$f'(x) = 2x, \qquad f''(x) = 2.$$

Hence

$$\boxed{\mathrm{d}(X_t^2) = (2X_t a_t + b_t^2)\,\mathrm{d}t + 2X_t b_t\,\mathrm{d}W_t.}$$

The term $b_t^2\,\mathrm{d}t$ is invisible to the classical chain rule.

Compensated square

Whenever the stochastic integral is integrable enough,

$$X_t^2 - X_0^2 - \int_0^t (2X_s a_s + b_s^2)\,\mathrm{d}s$$

is a martingale.

Exercise: arithmetic diffusion and its second moment

Let

$$X_t = x + \mu t + \sigma W_t.$$

Your turn

1. Compute $\mathrm{d}(X_t^2)$.
2. Find a compensated process with zero drift.
3. Use the result to calculate $\mathbb{E}[X_t^2]$.

Integrability check

Before taking expectations, explain why the stochastic integral has mean zero on $[0, T]$.

Exercise debrief: noise adds variance to the square

Since $\mathrm{d}X_t = \mu\,\mathrm{d}t + \sigma\,\mathrm{d}W_t$,

$$\mathrm{d}(X_t^2) = (2\mu X_t + \sigma^2)\,\mathrm{d}t + 2\sigma X_t\,\mathrm{d}W_t.$$

Thus

$$M_t = X_t^2 - x^2 - \int_0^t (2\mu X_s + \sigma^2)\,\mathrm{d}s$$

has zero drift. On bounded horizons, $\mathbb{E}\int_0^T X_s^2\,\mathrm{d}s < \infty$, so M is a true martingale. Consequently,

$$\boxed{\mathbb{E}[X_t^2] = (x + \mu t)^2 + \sigma^2 t.}$$

The answer is $(\text{mean})^2 + \text{variance}$, as it should be.

When may we erase the stochastic integral in expectation?

After applying Itô, the random term is often

$$\int_0^t H_s \, \mathrm{d}W_s, \qquad H_s = b_s f_x(s, X_s).$$

If

$$\mathbb{E}\left[\int_0^T H_s^2 \, \mathrm{d}s\right] < \infty,$$

then this is an L^2-martingale and its expectation is zero.

The important distinction

Pathwise local square-integrability generally gives only a *local* martingale. Zero drift alone does not prove the martingale property.

To exchange expectation and the drift integral, its absolute integrability must also be checked.

A reusable log rule for positive processes

Suppose $X_t > 0$ and

$$\frac{\mathrm{d}X_t}{X_t} = \alpha_t \,\mathrm{d}t + \beta_t \,\mathrm{d}W_t.$$

With $f(x) = \log x$,

$$\boxed{\mathrm{d}\log X_t = \left(\alpha_t - \frac{1}{2}\beta_t^2\right) \mathrm{d}t + \beta_t \,\mathrm{d}W_t.}$$

Volatility drag

The arithmetic growth rate is α_t, but the log growth rate is reduced by $\beta_t^2/2$.

The identity holds up to the first time X reaches zero if strict positivity is not known globally.

Exercise: invert an Ito process

Let $X_t \neq 0$ and

$$\mathrm{d}X_t = a_t\,\mathrm{d}t + b_t\,\mathrm{d}W_t.$$

Your turn

1. Use $f(x) = 1/x$ to find $\mathrm{d}(1/X_t)$.
2. Identify the term that the ordinary quotient rule would miss.
3. Specialize to $\mathrm{d}X_t/X_t = \mu\,\mathrm{d}t + \sigma\,\mathrm{d}W_t$.

Work only up to the first time X reaches zero.

Exercise debrief: inversion creates a drift correction

Since $f'(x) = -x^{-2}$ and $f''(x) = 2x^{-3}$,

$$\boxed{\mathrm{d}\left(\frac{1}{X_t}\right) = \left(-\frac{a_t}{X_t^2} + \frac{b_t^2}{X_t^3}\right)\mathrm{d}t - \frac{b_t}{X_t^2}\,\mathrm{d}W_t.}$$

For geometric dynamics, $R_t = 1/X_t$ therefore satisfies

$$\boxed{\frac{\mathrm{d}R_t}{R_t} = (\sigma^2 - \mu)\,\mathrm{d}t - \sigma\,\mathrm{d}W_t.}$$

The extra $+\sigma^2$ is the convexity correction. As a check, applying the product rule to $X_t R_t = 1$ makes every term cancel.

Two processes can share the same source of noise

Consider two Ito processes driven by the same Brownian motion:

$$\mathrm{d}X_t = a_t^X\,\mathrm{d}t + b_t^X\,\mathrm{d}W_t, \qquad \mathrm{d}Y_t = a_t^Y\,\mathrm{d}t + b_t^Y\,\mathrm{d}W_t.$$

Their infinitesimal product satisfies

$$\mathrm{d}X_t\,\mathrm{d}Y_t = b_t^X b_t^Y(\mathrm{d}W_t)^2 = b_t^X b_t^Y\,\mathrm{d}t.$$

Therefore,

$$\boxed{\mathrm{d}[X,Y]_t = b_t^X b_t^Y\,\mathrm{d}t.}$$

Interpretation

If two processes are exposed to the same random shock, their fluctuations create a non-zero cross-variation.

This cross term is what changes the usual multivariable chain rule.

Exercise: what survives in a two-dimensional expansion?

Suppose

$$\mathrm{d}X_t = a_t^X\,\mathrm{d}t + b_t^X\,\mathrm{d}W_t, \qquad \mathrm{d}Y_t = a_t^Y\,\mathrm{d}t + b_t^Y\,\mathrm{d}W_t.$$

Your turn

Using

$$(\mathrm{d}t)^2 = 0, \qquad \mathrm{d}t\,\mathrm{d}W_t = 0, \qquad (\mathrm{d}W_t)^2 = \mathrm{d}t,$$

simplify:

1. $(\mathrm{d}X_t)^2$,
2. $(\mathrm{d}Y_t)^2$,
3. $\mathrm{d}X_t\,\mathrm{d}Y_t$.

Question

Why does the product $\mathrm{d}X_t\,\mathrm{d}Y_t$ not disappear?

Ito's formula in two dimensions

Let

$$Z_t = f(t, X_t, Y_t), \qquad f \in C^{1,2},$$

where X and Y are continuous Ito processes. Then

$$\boxed{\begin{aligned} \mathrm{d}Z_t = {} & f_t \,\mathrm{d}t + f_x \,\mathrm{d}X_t + f_y \,\mathrm{d}Y_t \\ & + \frac{1}{2} f_{xx} \,\mathrm{d}[X]_t + f_{xy} \,\mathrm{d}[X, Y]_t + \frac{1}{2} f_{yy} \,\mathrm{d}[Y]_t. \end{aligned}}$$

The new ingredient

The mixed derivative f_{xy} is paired with the cross-variation $\mathrm{d}[X, Y]_t$.

This is the stochastic version of the second-order Taylor expansion in two variables.

Where does the mixed term come from?

For a small change $(\mathrm{d}X, \mathrm{d}Y)$, the second-order Taylor expansion gives

$$\begin{aligned} \mathrm{d}f \approx\ & f_x\,\mathrm{d}X + f_y\,\mathrm{d}Y \\ & + \frac{1}{2} f_{xx}(\mathrm{d}X)^2 + f_{xy}\,\mathrm{d}X\,\mathrm{d}Y + \frac{1}{2} f_{yy}(\mathrm{d}Y)^2. \end{aligned}$$

In ordinary calculus, all quadratic terms vanish at first order.

In stochastic calculus,

$$(\mathrm{d}W)^2 = \mathrm{d}t,$$

so quadratic terms involving Brownian increments survive.

Key idea

Ito's correction is simply the second-order Taylor expansion combined with the special scaling $\mathrm{d}W \sim \sqrt{\mathrm{d}t}$.

Exercise: derive the stochastic product rule

Take

$$f(x, y) = xy.$$

Then

$$f_x = y, \qquad f_y = x, \qquad f_{xx} = f_{yy} = 0, \qquad f_{xy} = 1.$$

Your turn

Apply the two-dimensional Ito formula to $X_t Y_t$. What extra term appears compared with the ordinary product rule?

Hint

Do not forget the mixed quadratic variation

$$\mathrm{d}[X, Y]_t.$$

Integration by parts has one extra term

Applying Ito's formula to $f(x, y) = xy$ gives

$$\boxed{\mathrm{d}(X_t Y_t) = X_t \,\mathrm{d}Y_t + Y_t \,\mathrm{d}X_t + \mathrm{d}[X, Y]_t.}$$

Equivalently,

$$\boxed{X_t Y_t = X_0 Y_0 + \int_0^t X_s \,\mathrm{d}Y_s + \int_0^t Y_s \,\mathrm{d}X_s + [X, Y]_t.}$$

Compare with ordinary calculus

The classical formula would contain only

$$\mathrm{d}(XY) = X \,\mathrm{d}Y + Y \,\mathrm{d}X.$$

The quadratic covariation is the stochastic correction.

Example: the product of two Ito processes

Suppose

$$\mathrm{d}X_t = a_t^X \,\mathrm{d}t + b_t^X \,\mathrm{d}W_t, \qquad \mathrm{d}Y_t = a_t^Y \,\mathrm{d}t + b_t^Y \,\mathrm{d}W_t.$$

Since

$$\mathrm{d}[X, Y]_t = b_t^X b_t^Y \,\mathrm{d}t,$$

integration by parts gives

$$\begin{aligned}\mathrm{d}(X_t Y_t) = &\left(X_t a_t^Y + Y_t a_t^X + b_t^X b_t^Y\right) \mathrm{d}t \\ &+ \left(X_t b_t^Y + Y_t b_t^X\right) \mathrm{d}W_t.\end{aligned}$$

What should you remember?

Multiplying two stochastic processes creates an additional drift term:

$$\boxed{b_t^X b_t^Y \,\mathrm{d}t.}$$

Independent Brownian noises have zero cross-variation

Now suppose

$$\mathrm{d}X_t = a_t^X \,\mathrm{d}t + b_t^X \,\mathrm{d}W_t^1, \qquad \mathrm{d}Y_t = a_t^Y \,\mathrm{d}t + b_t^Y \,\mathrm{d}W_t^2,$$

where W^1 and W^2 are independent Brownian motions.

Then

$$\mathrm{d}W_t^1 \,\mathrm{d}W_t^2 = 0,$$

and therefore

$$\boxed{\mathrm{d}[X,Y]_t = 0.}$$

Consequently

The product rule becomes

$$\mathrm{d}(X_t Y_t) = X_t \,\mathrm{d}Y_t + Y_t \,\mathrm{d}X_t.$$

There is no quadratic covariation correction.

Correlated Brownian motions create cross terms

Suppose instead

$$\mathrm{d}[W^1, W^2]_t = \rho \, \mathrm{d}t, \qquad -1 \le \rho \le 1.$$

If

$$\mathrm{d}X_t = a_t^X \, \mathrm{d}t + b_t^X \, \mathrm{d}W_t^1, \qquad \mathrm{d}Y_t = a_t^Y \, \mathrm{d}t + b_t^Y \, \mathrm{d}W_t^2,$$

then

$$\boxed{\mathrm{d}[X, Y]_t = \rho \, b_t^X b_t^Y \, \mathrm{d}t.}$$

Three useful cases

$\rho = 0$: no instantaneous covariance, $\rho = 1$: same shock, $\rho = -1$: opposite shocks.

Exercise: two correlated assets

Suppose

$$\frac{\mathrm{d}S_t^1}{S_t^1} = \mu_1\,\mathrm{d}t + \sigma_1\,\mathrm{d}W_t^1, \qquad \frac{\mathrm{d}S_t^2}{S_t^2} = \mu_2\,\mathrm{d}t + \sigma_2\,\mathrm{d}W_t^2,$$

with

$$\mathrm{d}[W^1, W^2]_t = \rho\,\mathrm{d}t.$$

Your turn

1. Compute $\mathrm{d}[S^1, S^2]_t$.
2. Use integration by parts to find $\mathrm{d}(S_t^1 S_t^2)$.
3. What happens to the drift when $\rho > 0$?

Financial interpretation

Correlation affects not only joint risk: through Ito's formula, it can also appear as a drift correction after a nonlinear transformation.

A multidimensional diffusion has vectors and matrices

Let

$$X_t = (X_t^1, \dots, X_t^d)^\top$$

satisfy

$$\boxed{\mathrm{d}X_t = b(t, X_t)\,\mathrm{d}t + \sigma(t, X_t)\,\mathrm{d}W_t,}$$

where

$$W_t \in \mathbb{R}^m, \qquad b(t,x) \in \mathbb{R}^d, \qquad \sigma(t,x) \in \mathbb{R}^{d\times m}.$$

Object	Dimension	Role
X_t	d	state variables
b	d	drift
W_t	m	independent shocks
σ	$d \times m$	exposure to shocks

The covariance matrix is $\sigma\sigma^\top$

From

$$\mathrm{d}X_t = b_t\,\mathrm{d}t + \sigma_t\,\mathrm{d}W_t,$$

the quadratic covariations satisfy

$$\boxed{\mathrm{d}[X^i, X^j]_t = (\sigma_t\sigma_t^\top)_{ij}\,\mathrm{d}t.}$$

Define the instantaneous covariance matrix

$$A_t = \sigma_t\sigma_t^\top.$$

Then

$$A_t^{ij}\,\mathrm{d}t$$

measures the instantaneous covariance between the i-th and j-th components of X.

Important

Even if the components of W are independent, the components of X need not be independent: the matrix σ can mix the shocks.

The multidimensional Ito formula

Let

$$\mathrm{d}X_t = b_t\,\mathrm{d}t + \sigma_t\,\mathrm{d}W_t, \qquad f : [0,T]\times\mathbb{R}^d \to \mathbb{R}.$$

For $f \in C^{1,2}$,

$$\boxed{\begin{aligned}\mathrm{d}f(t,X_t) = &\left(\partial_t f + b_t\cdot\nabla f + \frac{1}{2}\sum_{i,j=1}^{d}(\sigma_t\sigma_t^\top)_{ij}\partial_{ij}f\right)\mathrm{d}t\\ &+ \nabla f(t,X_t)^\top\sigma_t\,\mathrm{d}W_t.\end{aligned}}$$

Same structure as in dimension one

$$\text{time} + \text{drift}\times\text{slope} + \frac{1}{2}\,\text{covariance}\times\text{curvature} + \text{noise}.$$

Matrix notation makes Ito's correction compact

Let

$$A_t = \sigma_t \sigma_t^\top$$

and denote the Hessian of f by

$$D^2 f = \begin{pmatrix} \partial_{11} f & \cdots & \partial_{1d} f \\ \vdots & \ddots & \vdots \\ \partial_{d1} f & \cdots & \partial_{dd} f \end{pmatrix}.$$

Then

$$\sum_{i,j=1}^{d} A_t^{ij} \partial_{ij} f = \mathrm{Tr}\big(A_t D^2 f\big).$$

Hence

$$\boxed{\mathrm{d}f = \left(f_t + b_t \cdot \nabla f + \frac{1}{2} \mathrm{Tr}(A_t D^2 f) \right) \mathrm{d}t + \nabla f^\top \sigma_t \, \mathrm{d}W_t.}$$

The object to remember

In several dimensions,

Exercise: recover the two-dimensional formula

Let

$$X_t = \begin{pmatrix} X_t^1 \\ X_t^2 \end{pmatrix}, \qquad A_t = \begin{pmatrix} A_{11} & A_{12} \\ A_{12} & A_{22} \end{pmatrix}.$$

Your turn

Expand

$$\frac{1}{2}\,\mathrm{Tr}(A_t D^2 f).$$

Show that it equals

$$\frac{1}{2}A_{11}f_{11} + A_{12}f_{12} + \frac{1}{2}A_{22}f_{22}.$$

Why is there no $1/2$ in front of the mixed term?

The terms $(1,2)$ and $(2,1)$ both appear in the trace expansion, and $f_{12} = f_{21}$.

The generator is the drift produced by Ito's formula

For the diffusion

$$\mathrm{d}X_t = b(t, X_t)\,\mathrm{d}t + \sigma(t, X_t)\,\mathrm{d}W_t,$$

define

$$\boxed{\mathcal{L}f = b \cdot \nabla f + \frac{1}{2}\,\mathrm{Tr}\left(\sigma\sigma^\top D^2 f\right).}$$

Then Ito's formula becomes

$$\mathrm{d}f(t, X_t) = \left(\partial_t f + \mathcal{L}f\right)(t, X_t)\,\mathrm{d}t + \nabla f(t, X_t)^\top \sigma(t, X_t)\,\mathrm{d}W_t.$$

A useful consequence

If

$$\partial_t f + \mathcal{L}f = 0,$$

then $f(t, X_t)$ has zero drift and is a local martingale.

Transformations Solve Canonical SDEs

Geometric Brownian motion models relative shocks

Consider

$$\mathrm{d}S_t = \mu S_t \,\mathrm{d}t + \sigma S_t \,\mathrm{d}W_t, \qquad S_0 > 0, \quad \sigma > 0.$$

Over a short interval,

$$\frac{\Delta S_t}{S_t} \approx \mu \Delta t + \sigma \Delta W_t.$$

Thus the same Brownian shock creates a larger dollar move when the asset level is larger.

The useful transformation

The coefficients are multiplicative, so take a logarithm to turn the SDE into an additive one.

Ito solves geometric Brownian motion in one line

The log rule gives

$$\mathrm{d}\log S_t = \left(\mu - \frac{1}{2}\sigma^2\right)\mathrm{d}t + \sigma\,\mathrm{d}W_t.$$

Integrating from 0 to t,

$$\boxed{S_t = S_0 \exp\left\{\left(\mu - \frac{1}{2}\sigma^2\right)t + \sigma W_t\right\}.}$$

The explicit solution is strictly positive, so the logarithm was legitimate.

Law

$\log S_t$ is Gaussian; therefore S_t is lognormal.

Application: mean growth differs from typical growth

From the explicit GBM solution,

$$\mathbb{E}[S_t] = S_0 e^{\mu t}, \qquad \mathrm{Med}(S_t) = S_0 e^{(\mu - \sigma^2/2)t}.$$

For $\sigma > 0$,

$$\mathbb{P}(S_T > K) = \mathbb{P}\left(Z > \frac{\log(K/S_0) - (\mu - \sigma^2/2)T}{\sigma\sqrt{T}} \right), \quad Z \sim N(0,1).$$

Interpretation

Volatility leaves the mean growth μ unchanged but lowers the median log-growth. Rare large outcomes pull the mean upward.

Worked finance example: one year of GBM in numbers

Take

$$S_0 = 100, \qquad \mu = 8\%, \qquad \sigma = 20\%, \qquad T = 1.$$

The explicit solution gives

$$\mathbb{E}[S_1] = 100e^{0.08} \approx 108.33, \qquad \mathrm{Med}(S_1) = 100e^{0.06} \approx 106.18.$$

For a target level of 110,

$$\mathbb{P}(S_1 > 110) = 1 - \Phi\left(\frac{\log(1.10) - 0.06}{0.20}\right) \approx \boxed{43.0\%}.$$

Reading the numbers

The mean, median, and probability of beating a target answer three different questions. A positive expected return does not remove downside uncertainty.

Exercise: powers of a geometric Brownian motion

Let $p \in \mathbb{R}$ and $Y_t = S_t^p$.

Your turn

1. Find $\mathrm{d}Y_t/Y_t$.
2. Choose κ so that $e^{-\kappa t} S_t^p$ has zero drift.
3. Check your formula for $p = 0, 1, 2$.

The cases $p = -1$ and $p = 2$ are particularly useful consistency checks.

Exercise debrief: curvature changes the growth rate

Since $f'(s) = ps^{p-1}$ and $f''(s) = p(p-1)s^{p-2}$,

$$\boxed{\frac{\mathrm{d}(S_t^p)}{S_t^p} = \left[p\mu + \frac{1}{2}p(p-1)\sigma^2\right]\,\mathrm{d}t + p\sigma\,\mathrm{d}W_t.}$$

Hence

$$\kappa = p\mu + \frac{1}{2}p(p-1)\sigma^2.$$

For constant coefficients, $e^{-\kappa t}S_t^p$ is lognormal with constant mean and is a true martingale.

Why the correction changes sign

Powers are convex for $p > 1$ and concave for $0 < p < 1$; the Itô correction detects this curvature.

Time-dependent GBM uses the same transformation

Suppose

$$\frac{\mathrm{d}S_t}{S_t} = \mu_t \,\mathrm{d}t + \sigma_t \,\mathrm{d}W_t,$$

with progressively measurable coefficients satisfying

$$\int_0^T |\mu_t| \,\mathrm{d}t < \infty, \qquad \int_0^T \sigma_t^2 \,\mathrm{d}t < \infty \quad \text{a.s.}$$

Then

$$\boxed{S_t = S_0 \exp\left\{ \int_0^t \left(\mu_s - \frac{1}{2}\sigma_s^2 \right) \mathrm{d}s + \int_0^t \sigma_s \,\mathrm{d}W_s \right\}.}$$

If μ and σ are deterministic, the conditional law remains lognormal.

Mean reversion: the Ornstein–Uhlenbeck model

The SDE

$$\mathrm{d}X_t = \kappa(m - X_t)\,\mathrm{d}t + \sigma\,\mathrm{d}W_t, \qquad \kappa > 0,$$

has a restoring drift:

- above m, the drift is negative;
- below m, the drift is positive;
- $1/\kappa$ is the time scale of mean reversion.

It appears in short-rate models, spreads, temperatures, and other persistent but mean-reverting quantities.

A deterministic integrating factor removes the drift

Rewrite the SDE as

$$\mathrm{d}X_t + \kappa X_t \,\mathrm{d}t = \kappa m \,\mathrm{d}t + \sigma \,\mathrm{d}W_t.$$

Because $e^{\kappa t}$ has finite variation,

$$\mathrm{d}(e^{\kappa t} X_t) = e^{\kappa t} \,\mathrm{d}X_t + \kappa e^{\kappa t} X_t \,\mathrm{d}t,$$

with no covariation correction. Therefore

$$\mathrm{d}(e^{\kappa t} X_t) = \kappa m e^{\kappa t} \,\mathrm{d}t + \sigma e^{\kappa t} \,\mathrm{d}W_t.$$

Integrating and dividing by $e^{\kappa t}$ solves the SDE.

The OU solution displays memory loss explicitly

For $0 \le s < t$,

$$\boxed{X_t = m + (X_s - m)e^{-\kappa(t-s)} + \sigma \int_s^t e^{-\kappa(t-u)}\, \mathrm{d}W_u.}$$

Conditionally on $\mathcal{F}_s$,

$$\mathbb{E}[X_t \mid \mathcal{F}_s] = m + (X_s - m)e^{-\kappa(t-s)},$$

and

$$\mathrm{Var}(X_t \mid \mathcal{F}_s) = \frac{\sigma^2}{2\kappa}\left(1 - e^{-2\kappa(t-s)}\right).$$

The effect of the current deviation $X_s - m$ decays exponentially.

Exercise: read the long-run behavior from the formula

For the OU process, determine:

Your turn

1. the limiting conditional mean;
2. the limiting conditional variance;
3. the half-life of a deviation from m.

Then evaluate the half-life when $\kappa = 0.7$ per year.

Exercise debrief: mean reversion has a measurable speed

As $t - s \to \infty$,

$$\mathbb{E}[X_t \mid \mathcal{F}_s] \to m, \qquad \mathrm{Var}(X_t \mid \mathcal{F}_s) \to \frac{\sigma^2}{2\kappa}.$$

The equation $e^{-\kappa h} = 1/2$ gives the half-life

$$\boxed{h = \frac{\log 2}{\kappa}.}$$

For $\kappa = 0.7$,

$$h \approx 0.99 \text{ year}, \qquad e^{-0.7 \times 2} \approx 24.7\%.$$

After two years, only about one quarter of today's deviation from m remains in the conditional mean.

Exercise: obtain the CIR mean without solving the SDE

Assume a nonnegative CIR process satisfies

$$\mathrm{d}R_t = (\alpha - \beta R_t)\,\mathrm{d}t + \sigma\sqrt{R_t}\,\mathrm{d}W_t, \qquad R_0 = r_0, \quad \alpha, \beta, \sigma > 0.$$

Your turn

1. Compute $\mathrm{d}(e^{\beta t} R_t)$.
2. Integrate and take expectations.
3. Find $\mathbb{E}[R_t]$ and its long-run limit.

What part of the calculation uses the square-root volatility, and what part does not?

Exercise debrief: CIR and Vasicek share the same mean

With $X_t = e^{\beta t} R_t$, the product rule gives

$$\mathrm{d}X_t = \alpha e^{\beta t}\,\mathrm{d}t + \sigma e^{\beta t}\sqrt{R_t}\,\mathrm{d}W_t.$$

After localization and the usual integrability check, the stochastic integral has mean zero. Therefore

$$\mathbb{E}[X_t] = r_0 + \frac{\alpha}{\beta}(e^{\beta t} - 1),$$

and

$$\boxed{\mathbb{E}[R_t] = r_0 e^{-\beta t} + \frac{\alpha}{\beta}(1 - e^{-\beta t}) \longrightarrow \frac{\alpha}{\beta}.}$$

Interpretation

The diffusion coefficient controls dispersion, but the first moment follows the same mean-reversion ODE as the Vasicek model. The second moment is a natural next exercise.

Products, Ratios, and Several Risk Factors

The two-dimensional Ito formula includes a mixed term

For $f \in C^{1,2}([0,T] \times \mathbb{R}^2)$ and continuous Ito processes X, Y,

$$\begin{aligned} \mathrm{d}f(t, X_t, Y_t) = f_t \,\mathrm{d}t + f_x \,\mathrm{d}X_t + f_y \,\mathrm{d}Y_t \\ + \frac{1}{2} f_{xx} \,\mathrm{d}[X]_t + f_{xy} \,\mathrm{d}[X,Y]_t + \frac{1}{2} f_{yy} \,\mathrm{d}[Y]_t. \end{aligned}$$

If the driving Brownian motions satisfy $\mathrm{d}W_t \,\mathrm{d}B_t = \rho \,\mathrm{d}t$, the mixed term carries the correlation.

Memory aid

Diagonal second derivatives receive a factor $1/2$; the mixed derivative does not, because it appears twice in the Hessian sum.

The stochastic product rule has one extra term

Choosing $f(x, y) = xy$ gives

$$\boxed{\mathrm{d}(X_t Y_t) = X_t \,\mathrm{d}Y_t + Y_t \,\mathrm{d}X_t + \mathrm{d}[X, Y]_t.}$$

For

$$\mathrm{d}X_t = a_t \,\mathrm{d}t + b_t \,\mathrm{d}W_t, \qquad \mathrm{d}Y_t = c_t \,\mathrm{d}t + d_t \,\mathrm{d}B_t,$$

with $\mathrm{d}W_t \,\mathrm{d}B_t = \rho \,\mathrm{d}t$,

$$\mathrm{d}[X, Y]_t = \rho b_t d_t \,\mathrm{d}t.$$

This correction is the continuous-time record of simultaneous random shocks.

Application: the product of two correlated assets

Suppose

$$\frac{\mathrm{d}S_t}{S_t} = \mu_S\,\mathrm{d}t + \sigma_S\,\mathrm{d}W_t, \qquad \frac{\mathrm{d}V_t}{V_t} = \mu_V\,\mathrm{d}t + \sigma_V\,\mathrm{d}B_t,$$

and $\mathrm{d}W_t\,\mathrm{d}B_t = \rho\,\mathrm{d}t$. Then

$$\boxed{\frac{\mathrm{d}(S_tV_t)}{S_tV_t} = (\mu_S + \mu_V + \rho\sigma_S\sigma_V)\,\mathrm{d}t + \sigma_S\,\mathrm{d}W_t + \sigma_V\,\mathrm{d}B_t.}$$

Positive correlation increases the product's drift through covariation; negative correlation decreases it.

Exercise: normalize the sum of two Brownian motions

Let W, B be jointly Brownian with constant correlation $\rho > -1$.

Your turn

1. Compute $\mathrm{d}[(W + B)]_t$.
2. Find a constant c such that $c(W + B)$ has quadratic variation t.
3. What happens when $\rho = -1$?

Exercise debrief: correlation changes total variance

Using the product table,

$$\mathrm{d}[W+B]_t = (\,\mathrm{d}W_t + \,\mathrm{d}B_t)^2 = 2(1+\rho)\,\mathrm{d}t.$$

Therefore

$$\boxed{\widetilde{W}_t = \frac{W_t + B_t}{\sqrt{2(1+\rho)}}}$$

is a standard Brownian motion. This follows from joint Gaussianity, or from Levy's characterization. If $\rho = -1$, then $B = -W$ up to indistinguishability and the sum is identically zero.

Exercise: derive the ratio of two risky assets

With the same dynamics for S and V, let $R_t = S_t/V_t$.

Your turn

Use either the two-dimensional Itô formula or the product rule applied to $S_t(1/V_t)$. Find $\mathrm{d}R_t/R_t$.

Common trap

The ordinary quotient rule misses both the inverse correction $\sigma_V^2\,\mathrm{d}t$ and the cross correction $-\rho\sigma_S\sigma_V\,\mathrm{d}t$.

Exercise debrief: the denominator contributes convexity

The ratio satisfies

$$\boxed{\frac{\mathrm{d}R_t}{R_t} = \left(\mu_S - \mu_V + \sigma_V^2 - \rho\sigma_S\sigma_V\right)\,\mathrm{d}t + \sigma_S\,\mathrm{d}W_t - \sigma_V\,\mathrm{d}B_t.}$$

Sanity checks:

- if $S = V$, then $\rho = 1$, $\mu_S = \mu_V$, $\sigma_S = \sigma_V$, and every term vanishes;
- if V is deterministic, $\sigma_V = 0$, and ordinary quotient calculus returns.

Application: correlation determines a minimum-variance hedge

Consider the instantaneous shock

$$\sigma_X \, \mathrm{d}W_t - h\sigma_Y \, \mathrm{d}B_t, \qquad \mathrm{d}W_t \, \mathrm{d}B_t = \rho \, \mathrm{d}t.$$

Its variance rate is

$$q(h) = \sigma_X^2 + h^2\sigma_Y^2 - 2h\rho\sigma_X\sigma_Y.$$

Minimizing this quadratic gives

$$\boxed{h^* = \rho\frac{\sigma_X}{\sigma_Y}, \qquad q(h^*) = \sigma_X^2(1-\rho^2).}$$

Only the correlated component can be hedged; orthogonal noise remains.

Numerical check

If $\sigma_X = 25\%$, $\sigma_Y = 20\%$, and $\rho = 0.8$, then

$$h^* = 1, \qquad \sqrt{q(h^*)} = 25\%\sqrt{1-0.8^2} = 15\%.$$

Optional extension: rotating shocks preserves Brownian risk

Let W be a standard d-dimensional Brownian motion and let H_t be predictable with

$$H_t H_t^\top = I_d.$$

Define

$$B_t = \int_0^t H_s \, \mathrm{d}W_s.$$

Then B is a continuous local martingale and

$$\mathrm{d}[B^i, B^j]_t = (H_t H_t^\top)_{ij} \, \mathrm{d}t = \delta_{ij} \, \mathrm{d}t.$$

By Levy's characterization, B is again a standard d-dimensional Brownian motion.

One-dimensional surprise

$\int_0^t \operatorname{sgn}(W_s) \, \mathrm{d}W_s$ is Brownian, although its integrand depends on the path.

Finance Applications of Ito Calculus

Exercise: what does discounting do under the physical measure?

Let the bank account and asset satisfy

$$\mathrm{d}A_t = rA_t\,\mathrm{d}t, \qquad \mathrm{d}S_t = \mu S_t\,\mathrm{d}t + \sigma S_t\,\mathrm{d}W_t^{\mathbb{P}}.$$

Set $\widetilde{S}_t = S_t/A_t$.

Your turn

Find $\mathrm{d}\widetilde{S}_t/\widetilde{S}_t$. Does discounting by the bank account automatically make $\widetilde{S}$ a martingale under $\mathbb{P}$?

Hint: $A_t^{-1} = A_0^{-1}e^{-rt}$ has finite variation.

Exercise debrief: the measure matters

The product rule gives

$$\boxed{\frac{\mathrm{d}\widetilde{S}_t}{\widetilde{S}_t} = (\mu - r)\,\mathrm{d}t + \sigma\,\mathrm{d}W_t^{\mathbb{P}}.}$$

Under the physical measure $\mathbb{P}$, the drift generally remains $\mu - r$. Under a pricing measure $\mathbb{Q}$, the model is written

$$\frac{\mathrm{d}S_t}{S_t} = r\,\mathrm{d}t + \sigma\,\mathrm{d}W_t^{\mathbb{Q}},$$

so $\widetilde{S}$ has zero drift and is a local martingale. Standard integrability conditions make it a true martingale.

Ito turns an option value into drift plus hedgeable risk

Suppose $V_t = v(t, S_t)$, with

$$\mathrm{d}S_t = \mu S_t \,\mathrm{d}t + \sigma S_t \,\mathrm{d}W_t.$$

Then

$$\boxed{\mathrm{d}V_t = \left(v_t + \mu S_t v_s + \frac{1}{2}\sigma^2 S_t^2 v_{ss} \right) \mathrm{d}t + \sigma S_t v_s \,\mathrm{d}W_t.}$$

The derivatives have financial names:

$$\Delta = v_s, \qquad \Gamma = v_{ss}.$$

Delta controls first-order exposure; gamma measures curvature and creates the $\mathrm{It}\widehat{\mathrm{o}}$ correction.

Delta hedging cancels the instantaneous Brownian shock

Let $\Delta_t = v_s(t, S_t)$, and let the cash account finance every change in the stock position. The self-financing hedged *gain* is

$$\mathrm{d}\Pi_t := \mathrm{d}V_t - \Delta_t \,\mathrm{d}S_t.$$

Substitution gives

$$\begin{aligned}\mathrm{d}\Pi_t &= \left(v_t + \mu S_t v_s + \frac{1}{2}\sigma^2 S_t^2 v_{ss}\right)\mathrm{d}t + \sigma S_t v_s \,\mathrm{d}W_t \\ &\quad - v_s\left(\mu S_t \,\mathrm{d}t + \sigma S_t \,\mathrm{d}W_t\right) \\ &= \boxed{\left(v_t + \frac{1}{2}\sigma^2 S_t^2 v_{ss}\right)\mathrm{d}t.}\end{aligned}$$

The unknown physical drift μ cancels together with the random shock.

Bookkeeping point

This is the gain of a dynamically financed strategy. It is not the algebraic identity $\mathrm{d}(V_t - \Delta_t S_t) = \mathrm{d}V_t - \Delta_t \,\mathrm{d}S_t$.

Application: curvature creates deterministic hedged P&L

Take the toy claim $V_t = S_t^2$. Then

$$\Delta_t = 2S_t, \qquad \Gamma_t = 2.$$

The Itô formula gives

$$\mathrm{d}(S_t^2) = 2S_t\,\mathrm{d}S_t + \sigma^2 S_t^2\,\mathrm{d}t.$$

After shorting $2S_t$ shares,

$$\boxed{\mathrm{d}V_t - 2S_t\,\mathrm{d}S_t = \sigma^2 S_t^2\,\mathrm{d}t.}$$

Continuous rebalancing converts quadratic price variation into a finite-variation term.

One-day numerical scale

For an option with $\Gamma = 0.02$, $S = 100$, $\sigma = 20\%$, and $\Delta t = 1/252$, the gamma contribution is

$$\frac{1}{2}\Gamma\sigma^2 S^2 \Delta t \approx \$0.016$$

per option, before theta and financing.

Preview for Session 5: the Black–Scholes PDE

If the delta-hedged position is self-financing and locally riskless, it must earn the bank-account rate on its value $V_t - \Delta_t S_t$:

$$\mathrm{d}V_t - \Delta_t \,\mathrm{d}S_t = r(V_t - \Delta_t S_t)\,\mathrm{d}t.$$

Substituting $\Delta = v_s$ gives

$$\boxed{v_t + \frac{1}{2}\sigma^2 s^2 v_{ss} + r s v_s - r v = 0.}$$

Preview

Quadratic variation produces the gamma term; no-arbitrage determines how that locally riskless term must grow.

Exercise: convexity, concavity, and the Ito correction

Let $\mathrm{d}X_t = a_t\,\mathrm{d}t + b_t\,\mathrm{d}W_t$ and let f have no explicit time dependence.

Your turn

Compare the drift of $f(X_t)$ with the first-order prediction $f'(X_t)a_t\,\mathrm{d}t$ when:

1. f is convex;
2. f is concave;
3. f is linear.

Use $f(x) = x^2$ and $f(x) = \log x$ as examples.

Exercise debrief: volatility interacts with curvature

The correction is

$$\frac{1}{2} b_t^2 f''(X_t)\,\mathrm{d}t.$$

Therefore:

Shape of f	Sign of f''	Volatility correction
Convex	$f'' \geq 0$	nonnegative
Linear	$f'' = 0$	zero
Concave	$f'' \leq 0$	nonpositive

Thus volatility raises the drift of a convex transform and lowers that of a concave transform. This local fact is a cornerstone of option convexity.

Application: log quadratic variation is realized variance

For a positive asset with

$$\frac{\mathrm{d}S_t}{S_t} = \mu_t\,\mathrm{d}t + \sigma_t\,\mathrm{d}W_t, \qquad \mathrm{d}\log S_t = \left(\mu_t - \frac{1}{2}\sigma_t^2\right)\mathrm{d}t + \sigma_t\,\mathrm{d}W_t.$$

Therefore

$$\boxed{[\log S]_T = \int_0^T \sigma_t^2\,\mathrm{d}t.}$$

On a fine grid, squared log returns estimate this integrated variance.

Four-day calculation

For log returns $1.0\%, -0.5\%, 0.7\%, -0.2\%$,

$$\widehat{RV} = 0.01^2 + (-0.005)^2 + 0.007^2 + (-0.002)^2 = 1.78 \times 10^{-4}, \qquad \widehat{\sigma}_{\mathrm{ann}} = \sqrt{\frac{252}{4}\widehat{RV}} \approx 10.6\%.$$

Guided Practice and Common Traps

Rapid practice I: three transformations

Compute the differential of each process.

1. $Y_t = e^{\alpha t + \beta W_t}$.
2. $Z_t = W_t B_t$, where $\mathrm{d}W_t \, \mathrm{d}B_t = \rho \, \mathrm{d}t$.
3. Use $\widehat{\mathrm{Itô}}$ on e^{W_t} to remove $\mathrm{d}W$ from $\int_0^t e^{W_s} \, \mathrm{d}W_s$.

Work pattern

Write the relevant derivatives, substitute the dynamics, use the product table, and finally group drift and diffusion.

Rapid practice I: answers

$$\mathrm{d}Y_t = Y_t\left(\alpha + \frac{1}{2}\beta^2\right)\mathrm{d}t + \beta Y_t\,\mathrm{d}W_t,$$

$$\mathrm{d}Z_t = W_t\,\mathrm{d}B_t + B_t\,\mathrm{d}W_t + \rho\,\mathrm{d}t,$$

$$\int_0^t e^{W_s}\,\mathrm{d}W_s = e^{W_t} - 1 - \frac{1}{2}\int_0^t e^{W_s}\,\mathrm{d}s.$$

Checks:

- the exponential is convex, so its correction is positive;
- correlation contributes finite variation to the product;
- integrating $\mathrm{d}(e^{W_t})$ isolates the stochastic integral.

Rapid practice II: manufacture a martingale

Let

$$M_t = e^{at}\cos(bW_t), \qquad a, b \in \mathbb{R}.$$

Your turn

Choose a so that M has zero drift. Then decide whether it is only a local martingale or a true martingale on $[0, T]$.

Do not stop after saying "the drift is zero." Check the stochastic integrand.

Rapid practice II: answer and integrability check

The Itô formula gives

$$\mathrm{d}M_t = e^{at}\left(a - \frac{1}{2}b^2\right)\cos(bW_t)\,\mathrm{d}t - be^{at}\sin(bW_t)\,\mathrm{d}W_t.$$

Thus $a = b^2/2$ kills the drift. On $[0, T]$,

$$\mathbb{E}\int_0^T b^2 e^{b^2 t}\sin^2(bW_t)\,\mathrm{d}t \le b^2\int_0^T e^{b^2 t}\,\mathrm{d}t < \infty.$$

Therefore the stochastic integral is in L^2, and M is a true martingale on every bounded horizon.

Error diagnosis: find every incorrect line

A student writes, for $\mathrm{d}X_t = a_t\,\mathrm{d}t + b_t\,\mathrm{d}W_t$,

$$\mathrm{d}f(t, X_t) = f_x\,\mathrm{d}X_t + \frac{1}{2} b_t f_{xx}\,\mathrm{d}t,$$

$$\mathrm{d}(W_t B_t) = W_t\,\mathrm{d}B_t + B_t\,\mathrm{d}W_t,$$

$$\text{zero drift} \Longrightarrow \text{true martingale},$$

$$\frac{\mathrm{d}(S_t/A_t)}{S_t/A_t} = \sigma\,\mathrm{d}W_t^{\mathbb{P}}.$$

Here $\mathrm{d}W_t\,\mathrm{d}B_t = \rho\,\mathrm{d}t$, $\mathrm{d}S_t/S_t = \mu\,\mathrm{d}t + \sigma\,\mathrm{d}W_t^{\mathbb{P}}$, and $\mathrm{d}A_t/A_t = r\,\mathrm{d}t$.

Your turn

Correct the four lines and name the missing hypothesis or term in each one.

Error diagnosis: corrections

The corrected statements are

$$\mathrm{d}f(t, X_t) = \left(f_t + a_t f_x + \frac{1}{2} b_t^2 f_{xx} \right) \mathrm{d}t + b_t f_x \, \mathrm{d}W_t,$$

$$\mathrm{d}(W_t B_t) = W_t \, \mathrm{d}B_t + B_t \, \mathrm{d}W_t + \rho \, \mathrm{d}t,$$

$$\text{zero drift} \Longrightarrow \text{local martingale first},$$

$$\frac{\mathrm{d}(S_t/A_t)}{S_t/A_t} = (\mu - r) \, \mathrm{d}t + \sigma \, \mathrm{d}W_t^{\mathbb{P}}.$$

To obtain a true martingale, verify integrability. Discounting kills the drift only after moving to a pricing measure under which the asset drift is r.

Optional preview: when the classical Ito formula is not enough

The formula used today assumes $f \in C^{1,2}$. It cannot be applied directly at the kink of

$$f(x) = |x| \qquad \text{or} \qquad f(s) = (s - K)^+.$$

For nonsmooth convex functions, a local-time term appears in the generalized formula.

Practical rule

Before differentiating, locate singularities and ask whether the process can reach them. If it can, smooth approximation or the Ito–Tanaka formula may be needed.

This matters for option payoffs even when the option value itself becomes smooth before maturity.

A compact checklist for every Ito calculation

1. Identify time and state variables; compute f_t, f_x, f_{xx}.
2. Write the driving dynamics and their covariations.
3. Apply the product table: $(\mathrm{d}W)^2 = \mathrm{d}t,\ \mathrm{d}W\,\mathrm{d}B = \rho\,\mathrm{d}t$.
4. Keep every term of order $\mathrm{d}t$; discard higher-order products.
5. Group the result into drift and diffusion.
6. Check the domain and regularity of f.
7. If claiming a martingale or taking expectations, verify integrability.
8. In finance, state the probability measure.

What to remember from Session 4

- Brownian quadratic variation makes stochastic calculus second order.
- Covariation turns correlation into a $\mathrm{d}t$-term.
- The Itô formula is a chain rule with a curvature correction.
- Transformations solve useful SDEs and moment equations for GBM, OU, and CIR.
- Products, ratios, discounting, and delta hedging all use the same small table.
- Zero drift means local martingale; suitable integrability upgrades the conclusion.

Core habit

Always separate drift, diffusion, covariation, and measure. Most mistakes disappear once those four ingredients are visible.